
Signed Error Transport for Spatial Reconstruction: Accuracy Gains and a Calibration Penalty

Anonymous manuscript

Abstract

Sequential spatial reconstruction requires measurements that correct a map, rather than merely reduce a model’s uncertainty. We introduce counterfactual absolute-risk reduction (CARR), which transports a cross-validated estimate of signed model error through the ordinary-Kriging update and integrates the resulting folded-normal loss. A spatial residual-scale model separates acquisition uncertainty from the predictive distribution returned to the user. All acquisition parameters follow one fitting procedure across five analytic benchmarks, with ten paired initial designs and an unchanged exponential-kernel Kriging predictor. At the largest budgets, CARR reduces absolute reconstruction error by 22.3–68.3% relative to the stronger variance-based or gradient-weighted baseline. The convergence curves nevertheless establish a twofold measurement saving on only the diffusion problem; the four two-dimensional problems improve too late to establish that saving at the tested budgets. Ablations show that spatial scale adaptation drives the discontinuous cases, whereas signed transport is especially useful for diffusion. Improved mean estimates do not imply improved distributions: concentrating measurements near sharp changes inflates a global Kriging scale, worsens CRPS on four functions, and increases diffusion percentage error. These results identify both the value and the limits of correcting acquisition under a fixed, misspecified surrogate.

1 Introduction

An autonomous scattering experiment does not generally end when it finds a large peak. It must recover how a measured response changes across a specimen, including smooth regions, transitions, and narrow features. Gaussian-process-driven experiments make this spatial reconstruction practical by selecting measurements from the evolving map [22, 25]. The same problem arises in computer experiments, where each function evaluation may require an expensive simulation [29]. In both settings, the scientific target is a domain-wide reconstruction, not the location of a single optimum.

Kriging provides a mean map and an uncertainty map [7, 28]. Measuring where the latter is largest is attractive because it needs no estimate of the unknown function’s error. Yet posterior variance answers a conditional question: where is a chosen probabilistic model uncertain? An exponential kernel can be uncertain in an unimportant empty region while being confidently wrong near an unresolved boundary. Gradient weighting redirects attention to sharp features [22, 23], but neither a large variance nor a large local gradient quantifies how the next observation will change error elsewhere in the map.

This distinction motivates a decision rule based on *error transport*. A new observation contributes an innovation at its measurement location; Kriging propagates that innovation through a known influence function. We estimate the sign and spatial scale of the current model error from cross-validated auxiliary predictors, then score the absolute reconstruction loss after that innovation has propagated. The auxiliary predictors guide measurement selection only. Every reported map and predictive interval still comes from the same ordinary-Kriging model used by the comparison methods.

This construction belongs to the established tradition of statistical active learning and integrated experimental design [3, 6, 20]. It is not a new justification for variance reduction. Its distinguishing element is a signed, spatially scaled error law evaluated through the update of a possibly different predictor. That separation makes it possible to investigate an acquisition correction without simultaneously changing the reconstruction model. It also exposes a tension: measurements that improve a misspecified model’s mean can degrade its probabilistic forecasts. We examine that tension through paired trajectories, acquisition ablations, an independent counterfactual ranking experiment, and a bound connecting sharp sampled changes to inflation of the fitted process scale.

2 Methodology

2.1 A common ordinary-Kriging reconstruction

Write all inputs in the unit cube $\mathcal{X} = [0, 1]^d$, and let $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ contain noiseless observations. We use correlation $k_a(x, x') = \exp(-a\|x - x'\|_2)$, an unknown constant mean, and numerical nugget $\nu = 10^{-8}$. The positive inverse scale is selected after every observation by

$$\hat{a} = \arg \min_{\log a \in [-7, 7]} \frac{1}{\binom{n}{2}} \sum_{i < j} \left[1 - e^{-a\|x_i - x_j\|_2} - \frac{(y_i - y_j)^2}{S_y^2} \right]^2, \quad (1)$$

$$S_y^2 = \max\{(\max_i y_i - \min_i y_i)^2, 10^{-20}\}.$$

This makes the normalization of the empirical variogram explicit. Bounded scalar minimization uses tolerance 10^{-5} in $\log a$. With $R_{ij} = k_a(x_i, x_j) + \nu \mathbf{1}_{i=j}$, $u = R^{-1}\mathbf{1}$, and $q = \mathbf{1}^T u$, define

$$\hat{\beta} = u^T y / q, \quad \hat{s}^2 = \max\left\{ \frac{(y - \hat{\beta}\mathbf{1})^T R^{-1} (y - \hat{\beta}\mathbf{1})}{n - 1}, 10^{-16} \right\},$$

$$m(z) = \hat{\beta} + k_z^T R^{-1} (y - \hat{\beta}\mathbf{1}), \quad C(z, x) = \hat{s}^2 \left[k(z, x) - k_z^T R^{-1} k_x + \frac{(1 - k_z^T u)(1 - k_x^T u)}{q} \right]. \quad (2)$$

Here $k_z = (k(x_i, z))_i$ and $v(z) = C(z, z)$. The distribution reported for every method is $\mathcal{N}(m(z), v(z))$; it is a plug-in latent predictive distribution, without hyperparameter uncertainty. The fvGP backend of the supplied gpCAM implementation provides the matrix factorization and base posterior moments. A subclass adds the standard unknown-mean correction in Eq. (2), and gpCAM’s ask maximizes each custom acquisition. We do not substitute an independent GP implementation.

2.2 A working error law and a counterfactual decision

Let $e(z) = f(z) - m(z)$. For the purpose of acquisition, posit the working law

$$e \sim \mathcal{GP}(b, Q), \quad b(z) = \tilde{m}(z) - m(z), \quad Q(z, x) = \sqrt{\lambda(z)\lambda(x)} C(z, x). \quad (3)$$

The guide $\tilde{m}$ and positive scale λ are fitted using only observed data, as specified below. Multiplication by the positive diagonal scale preserves positive semidefiniteness. Equation (3) is an acquisition model for surrogate error, not a claim that the true deterministic function is a draw from this process. In particular, Q is not used to score predictive calibration.

For a prospective noiseless observation at x , the fixed-kernel ordinary-Kriging mean update is

$$m_x^+(z) = m(z) + h_x(z)[f(x) - m(x)], \quad h_x(z) = \frac{C(z, x)}{v(x) + \nu \hat{s}^2}. \quad (4)$$

The nugget appears because it is part of the implemented update. Under the working error law, the error after this update has marginal mean and variance

$$b_x^+(z) = b(z) - h_x(z)b(x),$$

$$q_x^+(z) = Q(z, z) - 2h_x(z)Q(z, x) + h_x(z)^2 Q(x, x). \quad (5)$$

Define the folded-normal absolute loss

$$\psi(b, s) = s \sqrt{\frac{2}{\pi}} \exp\left(-\frac{b^2}{2s^2}\right) + b \left[2\Phi\left(\frac{b}{s}\right) - 1 \right], \quad \psi(b, 0) = |b|. \quad (6)$$

Counterfactual absolute-risk reduction (CARR) chooses x maximizing

$$\alpha_{\text{CARR}}(x) = \frac{1}{M} \sum_{j=1}^M \left[\psi \left(b(z_j), \sqrt{Q(z_j, z_j)} \right) - \psi \left(b_x^+(z_j), \sqrt{q_x^+(z_j)} \right) \right], \quad (7)$$

where the z_j are uniformly weighted, unlabelled integration points. No values $f(z_j)$ are requested. In finite precision, post-update variances are floored at 10^{-14} , and standard deviations passed to ψ at 10^{-15} .

Proposition 1 (One-step absolute risk). *For fixed current predictor, influence function, and joint Gaussian error law (3), Eq. (7) is the exact expected decrease of empirical integrated absolute error on $\{z_j\}_{j=1}^M$ after update (4).*

Proof. The updated error is $e(z) - h_x(z)e(x)$. A linear combination of jointly Gaussian variables is Gaussian with the moments in Eq. (5); its expected absolute value is Eq. (6). Summing marginal expectations proves the statement without an independence assumption between integration points. $\square$

This is a conditional decision identity, not a guarantee of improvement for the true function. Scores can be negative: an innovation believed to help at x may propagate with the wrong sign elsewhere. In the special case $b = 0$, constant λ , and vanishing nugget, Eq. (7) becomes a positive multiple of integrated posterior-standard-deviation reduction. Thus its zero-bias homogeneous limit is a classical uncertainty-reduction criterion. Actual experiments refit a after observing a point, which is not modeled in this one-step identity.

2.3 One fitting procedure for all functions

We fit the error law at $n = 10$ and then every ten observations. Three folds are assigned by arrival index modulo three. Each expert predicts the held-out fold; the Kriging expert uses the current full-data $\hat{a}$ held fixed within the folds. This is conditional cross-validation, not a fully nested estimate of out-of-sample risk. Six experts are available, all with the same settings on every function:

1. ordinary Kriging as above;
2. thin-plate radial-basis interpolation with smoothing 10^{-8} and at most 64 nearest training points;
3. 32 extremely randomized trees, minimum leaf size one and all input features [10];
4. an additive cubic B-spline model [14], with $K = \min(25, \max(4, \lfloor n/(3d) \rfloor))$ uniformly spaced knots, linear extrapolation, and ridge penalty 10^{-5} ;
5. a CART partition with leaf size at least $\max(d + 2, \lfloor n/30 \rfloor)$ and an affine ridge model of penalty 10^{-4} in each leaf, following the model-tree principle [27]; leaf coordinates are centered and scaled by their standard deviations, floored at 10^{-6} ;
6. the same thin-plate interpolator applied to $\text{asinh}(y/s_0)$, with s_0 equal to the median of $|y|$, floored at 10^{-8} , followed by the inverse transform; inverse-transform arguments are clipped to $[-30, 30]$ to guard floating-point overflow.

Let $P_{i\ell}$ be these out-of-fold predictions. Following stacked prediction [4, 32], fit nonnegative weights summing to one by

$$\hat{w} = \arg \min_{w \geq 0, \mathbf{1}^T w = 1} \frac{\|Pw - y\|_2^2}{n \max\{\text{Var}(y), 10^{-16}\}} + 10^{-6} \|w\|_2^2. \quad (8)$$

SLSQP starts from equal weights, uses tolerance 10^{-10} and at most 200 iterations; a failed fit raises an error. Experts are then refitted to all current observations. Between refreshes their weights and non-GP predictions remain fixed, while the GP component uses the updated m . Expert and fold random seeds are deterministic functions of the trajectory seed; no function identity enters fitting.

For residuals $r_i = y_i - (P\hat{w})_i$ and out-of-fold Kriging variances v_i^{cv} , form

$$t_i = \frac{r_i^2}{\max\{v_i^{\text{cv}}, 10^{-12} \max(\text{Var}(y), 10^{-16})\}},$$

$$\bar{\lambda} = \text{clip}\left(\frac{1}{n} \sum_i t_i, 10^{-6}, 10^6\right), \quad \tilde{t}_i = \text{clip}(t_i, 10^{-6}, 10^6). \quad (9)$$

For $k \in \{3, 8, 16, 32\}$ capped at $n - 1$ and $\eta \in \{0, 1/4, 1/2, 3/4, 1\}$, compute leave-one-point-out nearest-neighbor scale estimates

$$\lambda_{-i}^{k,\eta} = \max \left\{ (1 - \eta) \bar{\lambda} + \frac{\eta}{k} \sum_{j \in \mathcal{N}_k^{(-i)}(x_i)} \tilde{t}_j, 10^{-6} \right\}. \quad (10)$$

Choose the pair minimizing $n^{-1} \sum_i [\log \lambda_{-i}^{k,\eta} + \tilde{t}_i / \lambda_{-i}^{k,\eta}]$, then predict $\lambda(x)$ with the same neighbor-average mixture using all observed points. This is a Gaussian scale score; equal scores retain the first pair in ascending k, η order. The global scale and stacking weights are not refitted inside this second leave-one-point-out score, so it is a regularized scale-selection heuristic rather than an unbiased nested validation procedure. All candidate scores, selected settings, expert errors, and convergence records are saved.

2.4 Position relative to adaptive surrogate design

The use of observed residuals to correct uncertainty is established. MEPE combines a prediction-error estimate with Kriging uncertainty [18]; SFCVT uses cross-validation with geometric separation [1]; EIGF compares the predicted response with a nearby observed response [16]. Local approximation error also motivates LOLA–Voronoi [8] and the Taylor-expansion-based TEAD rule [21]. GUESS and the GALE framework bring gradients and uncertainty into global fitting [17]. Reviews document a wide variety of related exploitation–exploration balances [9, 19].

CARR differs from a local residual or disagreement score by evaluating the transported innovation $h_x(z)b(x)$ throughout the domain and retaining its sign relative to $b(z)$. It differs from a variance multiplier by allowing a spatially varying error covariance and a mismatch between that covariance and the actual update. Neither stacking nor cross-validation under misspecification is new [2]. The contribution is their particular integration with fixed-predictor absolute-risk reduction, together with an empirical test of which parts matter. We compare directly with EIGF and integrated variance and standard-deviation reduction, but do not claim an exhaustive experimental comparison with MEPE, SFCVT, TEAD, or GALE.

3 Experimental Setup

3.1 Functions, paired trajectories, and independent evaluation

The five functions are the piecewise-sinusoidal cases, smooth oscillatory function, piecewise-linear case, and three-dimensional Stokes–Einstein diffusion coefficient studied by Noack et al. [22]. Piecewise I and II use the published ordered conditions at $x = 30$ and radius 20 on $[0, 50]^2$, with different jump magnitudes. They share boundary locations; Piecewise II amplifies the contrast. The oscillatory function is $\sin(2x) + \sin(2y) + (x/5)^2 - (y/5)^2$ on $[-10, 10]^2$. The piecewise-linear function has branches separated at $x = 21.961143$ and $y = 25$ on $[0, 50]^2$. Diffusion is evaluated over radius $r \in [1, 100]$ nm, temperature $T \in [0, 100]$ °C, and glycerol mass fraction $C_m \in [0, 100]\%$, with Cheng’s viscosity relation [5]. We express diffusion in $\mu\text{m}^2/\text{s}$. Complete equations appear in the supplementary numerical specification.

Seeds $0, \dots, 9$ each generate ten uniform initial points; every method uses exactly these same points and responses for that seed. The observation budgets are 50, 100, 200, 500 for each two-dimensional function, with 1000 added for diffusion. Adaptive and random trajectories are nested across budgets. At each grid budget, the remaining $n - 10$ points are an independently ordered, uniformly thinned Cartesian lattice with $\lceil (n - 10)^{1/d} \rceil$ nodes per axis including endpoints. This yields a full-domain

grid at every budget; grids at different budgets are not nested. All methods update the same Kriging model after each individual observation.

Evaluation uses a fixed shifted Cartesian grid of $100^2 = 10,000$ points in two dimensions and $40^3 = 64,000$ in three. Axis offsets are $(0.371, 0.529, 0.683)$ divided by the corresponding grid side count. These evaluation points are independent of acquisition generation; the minimum distance from training points is checked and logged at every checkpoint. Their responses are computed only for scoring and never enter acquisition, guide fitting, or candidate generation. The headline error is

$$L_1(n) = N_{\text{test}}^{-1} \sum_j |m_n(z_j) - f(z_j)|. \quad (11)$$

The normalization is by integration-domain volume, implemented as an average, not by response amplitude. We also compute RMSE, Gaussian CRPS [11], and empirical coverage of central predictive intervals at nominal levels $0.1, 0.2, \dots, 0.9, 0.95, 0.99$ [12]. Diffusion MAPE is $100N_{\text{test}}^{-1} \sum_j |m_n(z_j) - f(z_j)|/|f(z_j)|$, without an artificial denominator floor. Every table reports sample standard deviations over ten seeds, not standard errors or confidence intervals.

3.2 Acquisition optimization and comparison criteria

Every adaptive method maximizes over the same geometry-based candidate-generation rule: 256 scrambled Sobol points [30], 16 random points on each domain face, all corners, and at most 64 randomly retained midpoints between existing points and their two nearest neighbors. Points within 10^{-7} of a previous observation are excluded. Candidate sets therefore differ after trajectories diverge, but their recipe, random seed rule, and maximum size are identical. Midpoints depend on locations, not responses. gpCAM evaluates the custom acquisition on this list and selects one maximizer. CARR uses 512 independent, unlabelled scrambled Sobol integration points, refreshed on the same ten-observation schedule as the guide. The oracle is called only after that single location has been selected.

SMART maximizes $v(x)$. Our normalized gSMART implementation maximizes

$$v(x) [1 + c_n g_n(x)^2], \quad g_n(x)^2 = \frac{\|\nabla m_n(x)\|_2^2}{\max(\text{Var}(y), 10^{-16})}, \quad (12)$$

$$c_n = \frac{1}{\max(\max_{x \in \mathcal{A}_n} g_n(x)^2, 10^{-16})}.$$

Gradients are analytic in unit-cube coordinates. This rule constrains the multiplier to $[1, 2]$ on the reference candidate set and is used unchanged for every function. It is a scale-normalized instance of the gradient-weighted SMART family, not a literal reconstruction of the original paper’s manually chosen coefficient. We call it gSMART in tables for brevity. An initially tried fixed coefficient caused clustering near jumps; the diagnostic history and unresolved reproduction differences are documented in Appendix A.

Three established acquisitions provide additional controls: EIGF, $(m(x) - y_{\text{nearest}(x)})^2 + v(x)$; integrated variance reduction (IVR), the average of $C(z, x)^2/[v(x) + \nu \hat{s}^2]$; and integrated standard-deviation reduction (ISR), the average of $\sqrt{v(z)} - \sqrt{v(z) - C(z, x)^2/[v(x) + \nu \hat{s}^2]}$. They use the same candidates and integration points where applicable. CARR ablations set $b = 0$ (zero bias), replace b by $|b|$ (unsigned bias), or force $\eta = 0$ (global scale). A simpler disagreement control scores $|\tilde{m}(x) - m(x)|\sqrt{v(x)}$. All ablations retain the common predictor and are run for all five functions with ten paired seeds; their maximum-budget results are emphasized in the paper.

3.3 Development, execution, and validation

The finite-pool optimizer, variogram normalization, expert library, and spatial scale model were developed in exploratory runs using seeds 0 and 1 on all five functions. Final reporting includes those two seeds as well as eight further seeds, rather than presenting the full sweep as a blind holdout. There is no function-specific selection rule, but the five-function benchmark was visible during method development. The evidence therefore supports this benchmark comparison, not a claim of generalization to unseen function families.

Table 1: Domain-averaged absolute error (mean $\pm$ sample standard deviation over ten paired seeds). Diffusion errors are in $\mu\text{m}^2/\text{s}$; the other rows retain their function units. Bold marks the lowest mean in each row. All reconstructions use the same ordinary-Kriging model.

Function	n	Grid	Random	SMART	gSMART	CARR
Piecewise I	50	8.557 ± 0.603	12.431 ± 1.525	8.443 ± 0.762	7.744 ± 0.521	8.497 ± 0.939
	100	5.462 ± 0.238	7.701 ± 0.771	5.478 ± 0.266	5.206 ± 0.250	5.074 ± 0.444
	200	3.910 ± 0.118	4.949 ± 0.310	3.733 ± 0.167	3.421 ± 0.192	3.051 ± 0.143
	500	2.302 ± 0.062	3.004 ± 0.206	2.299 ± 0.117	2.038 ± 0.066	1.583 ± 0.054
Piecewise II	50	24.635 ± 1.414	30.419 ± 2.824	25.184 ± 2.358	23.537 ± 1.004	24.688 ± 2.677
	100	16.411 ± 0.512	20.932 ± 1.777	16.864 ± 0.733	16.060 ± 0.614	15.947 ± 0.925
	200	12.335 ± 0.272	14.792 ± 1.051	11.858 ± 0.336	10.877 ± 0.411	9.559 ± 0.560
	500	7.503 ± 0.132	9.263 ± 0.486	7.208 ± 0.349	6.595 ± 0.198	4.751 ± 0.158
Smooth osc.	50	1.014 ± 0.027	0.956 ± 0.018	1.039 ± 0.054	1.008 ± 0.040	0.946 ± 0.040
	100	0.969 ± 0.013	0.848 ± 0.021	0.874 ± 0.031	0.877 ± 0.021	0.729 ± 0.024
	200	0.328 ± 0.010	0.675 ± 0.020	0.535 ± 0.012	0.577 ± 0.022	0.337 ± 0.014
	500	0.195 ± 0.004	0.347 ± 0.022	0.190 ± 0.004	0.234 ± 0.006	0.105 ± 0.002
Non-diff.	50	4.292 ± 0.330	4.347 ± 0.475	3.478 ± 0.350	3.271 ± 0.263	3.543 ± 0.458
	100	2.464 ± 0.136	2.857 ± 0.206	2.393 ± 0.148	2.198 ± 0.083	1.873 ± 0.221
	200	1.434 ± 0.020	2.006 ± 0.150	1.650 ± 0.081	1.541 ± 0.058	1.130 ± 0.063
	500	1.203 ± 0.026	1.268 ± 0.055	0.988 ± 0.025	0.899 ± 0.031	0.616 ± 0.033
3D diffusion	50	26.805 ± 3.518	4.204 ± 0.912	18.501 ± 3.897	13.717 ± 2.000	5.063 ± 1.677
	100	17.764 ± 1.507	3.585 ± 0.737	12.434 ± 1.823	9.774 ± 1.024	3.276 ± 0.914
	200	12.988 ± 0.559	2.903 ± 0.402	8.948 ± 0.839	7.722 ± 0.716	2.524 ± 0.370
	500	8.519 ± 0.117	2.140 ± 0.133	6.233 ± 0.386	5.135 ± 0.337	1.802 ± 0.097
	1000	6.105 ± 0.039	1.654 ± 0.149	4.593 ± 0.359	4.024 ± 0.197	1.275 ± 0.052

The implementation reuses the supplied gpCAM snapshot with fvGP 4.7.9. Bluesky’s orchestration and SciAnalysis’s diffraction reduction were inspected but are unnecessary for analytic scalar measurements. Numerical work uses NumPy, SciPy, and scikit-learn, and figures use Matplotlib [13, 15, 26, 31]. Experiments ran on CPU with one numerical-library thread per worker and at most 32 workers. Independent algebra checks compare the gpCAM adapter with direct ordinary-Kriging equations, including leave-one-out residuals; maximum discrepancies are below 4×10^{-14} . A 400,000-draw Monte Carlo check agrees with Eq. (7) within 1.4×10^{-4} in its tested two-variable example. Serial replay, separate from parallel experiment timing, measures decision cost and verifies the selected locations against saved trajectories.

4 Experimental Results

Table 1 and Figure 1 report every headline configuration. Relative to the better of SMART and gSMART at the largest budget, CARR reduces mean absolute error by 22.3% on Piecewise I, 28.0% on Piecewise II, 44.7% on the oscillatory case, 31.4% on the piecewise-linear case, and 68.3% on diffusion. These are matched-budget error reductions; they do not themselves establish the same fraction of measurements saved.

Piecewise I: a late crossover. CARR starts above both SMART and gSMART at 50 observations. Its mean moves slightly below them at 100, with overlapping standard-deviation bands, then separates at 200 and 500. The final error is 1.583 ± 0.054 , versus gSMART’s 2.038 ± 0.066 . The curved boundary and the smooth background both require coverage; an initially unreliable guide can spend points fitting a wrongly located transition. The spatial scale fit becomes useful once enough cross-validated residuals localize that transition. Nevertheless, CARR at 200 observations remains above the best baseline at 500.

Piecewise II: stronger contrast, the same delayed separation. CARR is again worse than gSMART at 50, only slightly better at 100, and clearly lower at 200 and 500. The final error is 4.751 ± 0.158 , versus 6.595 ± 0.198 . Larger jumps increase the value of resolving the boundary but also the penalty for transporting an innovation across it. The identical boundary geometry does

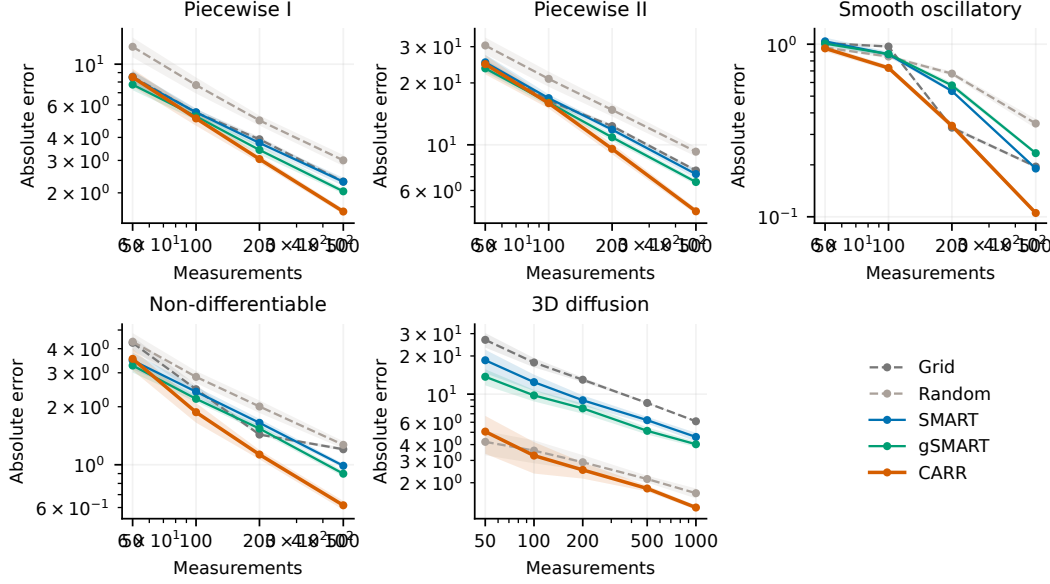


Figure 1: Absolute reconstruction error across the complete budget sweep. Lines are ten-seed means and bands are one sample standard deviation. Axes are logarithmic. CARR improves late-budget accuracy on every function, but its initial behavior and rate of separation vary substantially.

not make the early guide correct; the curve shows no demonstrated factor-of-two saving. The scale ablation below isolates why late boundary refinement succeeds.

Smooth oscillations: persistent mean improvement without a measured twofold saving. CARR’s mean is below both adaptive baselines at every budget; one-standard-deviation bands separate from 100 onward. At 200, however, the grid has lower mean error (0.328 ± 0.010 versus 0.337 ± 0.014), and CARR still has higher error than SMART at 500 (0.190 ± 0.004). At 500 CARR reaches 0.105 ± 0.002 . This function is additive and its oscillations occupy the whole domain. A learned additive guide can recognize that structure, but the returned exponential-kernel predictor must still acquire enough spatial resolution to interpolate it. The sampling grid is particularly competitive when its spacing resolves the oscillations.

Piecewise-linear response: recovering branches takes an initial investment. At 50, CARR is worse than SMART and gSMART. It becomes lower at 100 and maintains separated bands thereafter, reaching 0.616 ± 0.033 at 500 versus gSMART’s 0.899 ± 0.031 . Here a partition-linear guide can represent branch interiors efficiently once it learns the split. The acquisition must still localize the discontinuity for a continuous Kriging predictor; this is why its 200-observation error (1.130 ± 0.063) remains above the final baseline target. Faster terminal decay is not sufficient evidence of a twofold sample saving.

Diffusion: a large conditional saving, with a strong random baseline. CARR is below SMART and gSMART at every budget with separated bands throughout. Its 100-observation mean (3.276 ± 0.914) is already below the lower baseline mean at 1000 (4.024 ± 0.197). Thus the observed-budget saving is tenfold against these particular adaptive baselines. Random sampling is much stronger than those baselines on this function: it outperforms CARR at 50, remains close at 100 and 200, and reaches 1.654 ± 0.149 at 1000 compared with CARR’s 1.275 ± 0.052 . The large ratio against SMART/gSMART must therefore not be interpreted as a tenfold gain over good sampling strategies generally.

What the curves establish. Table 2 uses the same reference convention for every function: the lower SMART/gSMART mean at the maximum tested budget. A substantial saving requires reaching that target by half that budget and having nonoverlapping one-standard-deviation bands below both baselines on a majority of tested budgets. Only diffusion satisfies both conditions. The two-

Table 2: A fixed reference for sample efficiency. The target is the lower SMART/gSMART mean error at the maximum budget N . n_* is the first tested CARR budget reaching that target. Separation counts budgets where the CARR upper one-standard-deviation band is below both baseline lower bands. Unmeasured crossings are not interpolated.

Function	N	Target	n_*	N/n_*	Separated
Piecewise I	500	2.038	500	1	2/4
Piecewise II	500	6.595	500	1	2/4
Smooth osc.	500	0.190	500	1	3/4
Non-diff.	500	0.899	500	1	3/4
3D diffusion	1000	4.024	100	10	5/5

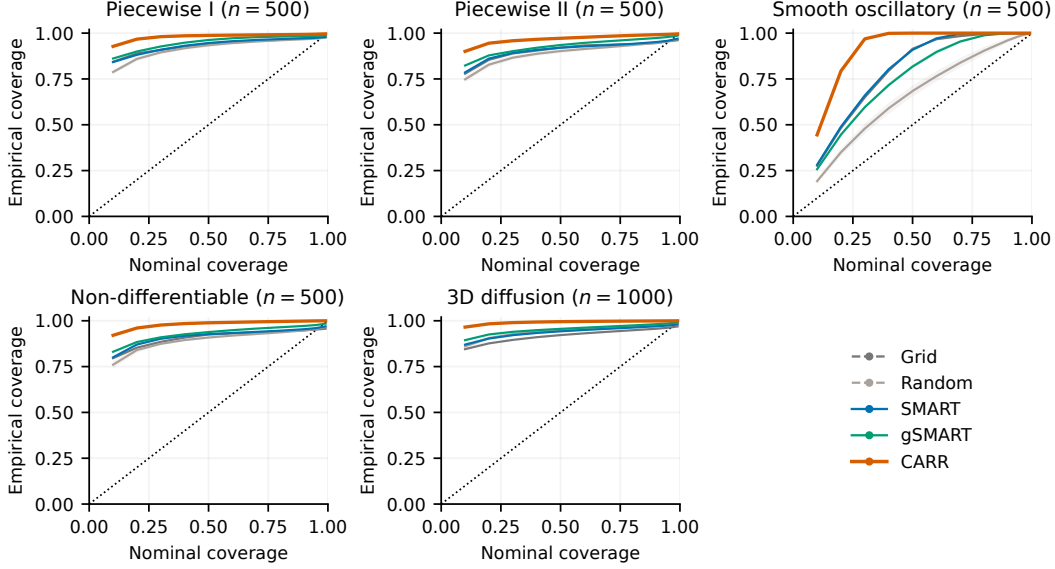


Figure 2: Coverage of the common Kriging predictive distribution at the largest budget for each function. The diagonal is ideal spatial coverage, and bands show one seed standard deviation. Curves above the diagonal indicate intervals that contain the truth too often for their nominal level. Full curves at every budget are provided in the numerical supplement.

dimensional targets are first crossed at the last tested budget. The coarse intervals between checkpoints do not locate the actual crossing times, and we do not interpolate them into favorable efficiency ratios. In particular, the evidence does not establish substantial savings on four of the five functions.

Baseline reproduction has a material limit. The baseline mean curves all improve with budget, and the adaptive criteria outperform random sampling on the three discontinuous functions. They do not reproduce universal dominance over both random and grid. Full-domain grids are competitive on the two-dimensional problems, while random sampling is markedly better on diffusion. Appendix A records changes to the variogram scaling, gradient computation, gradient coefficient, and continuous optimizer that were investigated. These discrepancies remain unresolved as a reproduction of the original qualitative ranking. The comparisons here are controlled comparisons of the stated gpCAM implementations; the diffusion speedup is conditional on that configuration.

Point accuracy and distribution quality diverge. Supplementary Table 8 supplies RMSE, CRPS, and coverage for every method and all 21 configurations. CARR also has lower final RMSE than SMART and gSMART on all functions, but worse final CRPS on four of them. For Piecewise I, its CRPS is 9.593 compared with gSMART’s 4.860; for Piecewise II it is 21.696 versus 12.550; for the piecewise-linear case it is 4.524 versus 1.728. On diffusion it is 20.897 versus 11.147, although the mean prediction is much more accurate. Only the smooth oscillatory case improves CRPS against

Table 3: Diffusion MAPE (percent, ten-seed mean $\pm$ standard deviation). All predictions remain on the original response scale.

n	Grid	Random	SMART	gSMART	CARR
50	4826.4 $\pm$ 3652.4	1328.1 $\pm$ 562.5	2030.2 $\pm$ 545.0	1934.3 $\pm$ 418.2	2515.7 $\pm$ 1503.8
100	1217.1 $\pm$ 226.9	830.3 $\pm$ 525.6	953.8 $\pm$ 135.5	1012.6 $\pm$ 101.5	2489.5 $\pm$ 819.0
200	728.1 $\pm$ 153.7	728.0 $\pm$ 276.2	587.5 $\pm$ 68.6	631.5 $\pm$ 44.2	1695.7 $\pm$ 539.1
500	414.5 $\pm$ 28.8	408.7 $\pm$ 166.1	340.2 $\pm$ 29.9	364.0 $\pm$ 42.1	1208.9 $\pm$ 274.4
1000	277.2 $\pm$ 5.3	268.3 $\pm$ 60.6	214.1 $\pm$ 12.1	231.0 $\pm$ 16.5	796.7 $\pm$ 150.5

Table 4: Acquisition controls at the largest budget for each function (ten seeds, mean $\pm$ standard deviation). All methods return the same Kriging predictor. Full trajectories and all predictive metrics are supplied in the machine-readable results.

Method	Piecewise I	Piecewise II	Smooth osc.	Non-diff.	Diffusion
CARR	1.583 $\pm$ 0.054	4.751 $\pm$ 0.158	0.105 $\pm$ 0.002	0.616 $\pm$ 0.033	1.275 $\pm$ 0.052
Zero bias	1.512 $\pm$ 0.054	5.131 $\pm$ 0.117	0.222 $\pm$ 0.009	0.886 $\pm$ 0.137	1.862 $\pm$ 0.113
Unsigned bias	1.562 $\pm$ 0.056	4.756 $\pm$ 0.116	0.103 $\pm$ 0.002	0.624 $\pm$ 0.033	1.528 $\pm$ 0.033
Global scale	2.230 $\pm$ 0.179	6.667 $\pm$ 0.512	0.104 $\pm$ 0.001	0.647 $\pm$ 0.050	1.546 $\pm$ 0.068
Disagreement	1.649 $\pm$ 0.090	5.553 $\pm$ 0.252	0.096 $\pm$ 0.001	1.117 $\pm$ 0.046	3.043 $\pm$ 0.102
ISR	2.408 $\pm$ 0.124	7.516 $\pm$ 0.282	0.218 $\pm$ 0.003	1.042 $\pm$ 0.063	2.754 $\pm$ 0.344
IVR	2.382 $\pm$ 0.093	7.562 $\pm$ 0.299	0.211 $\pm$ 0.006	1.044 $\pm$ 0.024	2.825 $\pm$ 0.248
EIGF	2.159 $\pm$ 0.076	7.200 $\pm$ 0.230	0.376 $\pm$ 0.013	0.912 $\pm$ 0.025	3.420 $\pm$ 0.076

both (0.169 versus SMART’s 0.176). Figure 2 shows pervasive overcoverage. CARR’s nominal 90% intervals cover 99.3%, 99.2%, 100.0%, 99.8%, and 99.9% of test points, respectively.

The objective mismatch is especially clear for diffusion (Table 3). At 1000 observations, CARR’s MAPE is 796.7%, compared with 214.1% for SMART and 268.3% for random sampling. Percentage error emphasizes very small responses, whereas absolute risk emphasizes large contributions to the volume integral. CARR does not enforce positive predictions or optimize logarithmic error; near small true diffusivities, even small absolute errors produce large percentage errors. Its favorable absolute-error curve must be read together with this regression.

5 Analysis

5.1 Mechanism analysis: spatial scale and signed transport have different roles

Table 4 isolates acquisition changes under an identical predictor. Replacing the learned spatial scale by a global scalar raises Piecewise I error from 1.583 to 2.230 and Piecewise II from 4.751 to 6.667. In contrast, removing the signed bias altogether improves Piecewise I to 1.512. There, local residual scale is more useful than an imperfect signed guide. On Piecewise II, unsigned transport is essentially as accurate in mean as signed transport (4.756 versus 4.751), while zero bias is worse (5.131). The evidence supports using local model-error information, but not a universal benefit from its sign on discontinuous functions.

The smooth function provides a different decomposition. Zero bias degrades error to 0.222, compared with 0.105 for full CARR, while global scaling and unsigned bias give 0.104 and 0.103. A signed guide is therefore unnecessary for most of this gain; recognizing broad systematic error is sufficient. Indeed, the cheaper local disagreement criterion achieves 0.096, below full CARR. At the final refresh the additive-spline expert receives mean weight 0.999, consistent with the function’s additive structure. This favorable match between library and benchmark is a limitation of any generality claim.

For the piecewise-linear function, the partition-linear expert receives mean weight 0.890. Its guide reduces the burden of representing affine interiors, while the acquisition refines the remaining boundaries. Zero bias gives 0.886 and local disagreement 1.117, compared with 0.616 for CARR. Unsigned bias remains close at 0.624. For diffusion, signed transport is more consequential: error rises from 1.275 to 1.528 with unsigned bias, 1.862 with zero bias, and 3.043 with local disagreement.

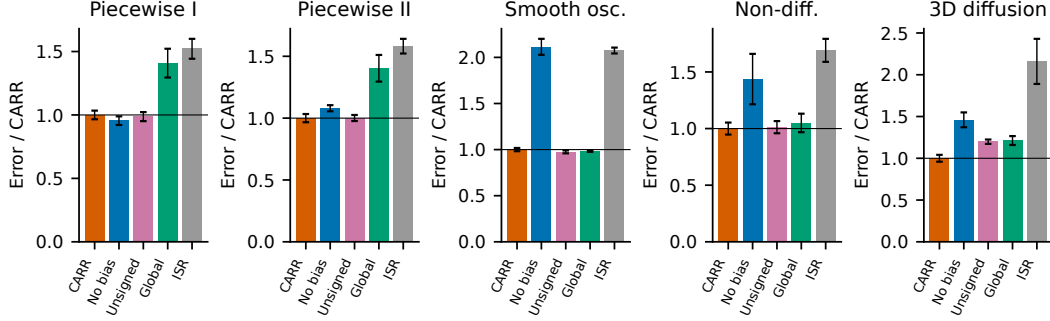


Figure 3: Selected acquisition ablations at the maximum budget. Errors are divided by the CARR mean for each function; whiskers are the corresponding seed standard deviations on that same scale. The zero-bias and unsigned variants show that the value of signed transport depends on function structure. Table 4 includes all controls in original units.

Table 5: Counterfactual ranking audit at $n = 200$: Spearman correlation between predicted scores and realized one-step absolute-error reduction, mean $\pm$ standard deviation across ten seeds. Each seed uses 64 independent candidate locations and the same CARR history for every score.

Function	CARR	Zero bias	Unsigned bias	gSMART
Piecewise I	0.246 ± 0.123	0.298 ± 0.138	0.286 ± 0.139	0.347 ± 0.120
Piecewise II	0.306 ± 0.102	0.293 ± 0.097	0.277 ± 0.096	0.301 ± 0.080
Smooth osc.	0.705 ± 0.078	-0.142 ± 0.125	0.337 ± 0.095	-0.112 ± 0.110
Non-diff.	0.441 ± 0.238	0.186 ± 0.170	0.356 ± 0.188	0.130 ± 0.135
3D diffusion	0.488 ± 0.143	0.169 ± 0.175	0.418 ± 0.108	0.014 ± 0.124

Full CARR is also lower than IVR (2.825), ISR (2.754), and EIGF (3.420). An innovation in a high-diffusivity region can harm many lower-diffusivity locations if propagated in the wrong direction; the counterfactual domain integral explicitly accounts for this effect.

The spatial evidence is consistent with this account. In the normalized band of width 0.025 around the Piecewise I boundaries, final mean absolute error is 12.351 for CARR versus 16.941 for gSMART; in the remaining bulk it is 0.627 versus 0.716. For Piecewise II the corresponding boundary errors are 37.590 and 56.486, with bulk errors 1.837 and 2.168. For the piecewise-linear case, the boundary error falls from 6.483 to 3.610, while bulk error changes little (0.295 to 0.293). These regions are defined for retrospective diagnosis only; boundary equations are never supplied to the acquisition.

Diffusion makes boundary placement especially consequential because $D \propto 1/r$. At 1000 observations, 42.2% of SMART points and 24.0% of CARR points lie exactly on some domain face. CARR focuses its boundary effort rather than uniformly favoring edges: it puts more points on the minimum-radius face and fewer on all faces combined. In the lowest 5% of the radius interval, CARR’s mean error is 12.422 versus gSMART’s 43.266; outside that region it is 0.688 versus 1.958. Random sampling avoids face concentrations entirely and consequently provides a much stronger comparator here than SMART. These are associations within the observed designs, not a separately randomized test of candidate-face sampling.

Testing the counterfactual score directly. We freeze each of the ten CARR histories at 200 observations and construct 64 new independent Sobol candidates per function and seed. All acquisition scores are calculated before any candidate response is evaluated. We then evaluate those responses retrospectively and compute the actual change in dense-grid absolute error under the fixed-kernel mean update. These extra evaluations are used only in this offline audit; they are not measurements in the reported learning curves. A fresh gpCAM fit at fixed α verifies the rank-one update numerically.

Table 5 reports seed-level Spearman correlations, avoiding the treatment of correlated candidates as independent experimental repeats. CARR predicts realized gains substantially better than gSMART on oscillations (0.705 versus -0.112), the piecewise-linear function (0.441 versus 0.130), and diffusion (0.488 versus 0.014). On Piecewise I it is worse (0.246 versus 0.347), and on Piecewise

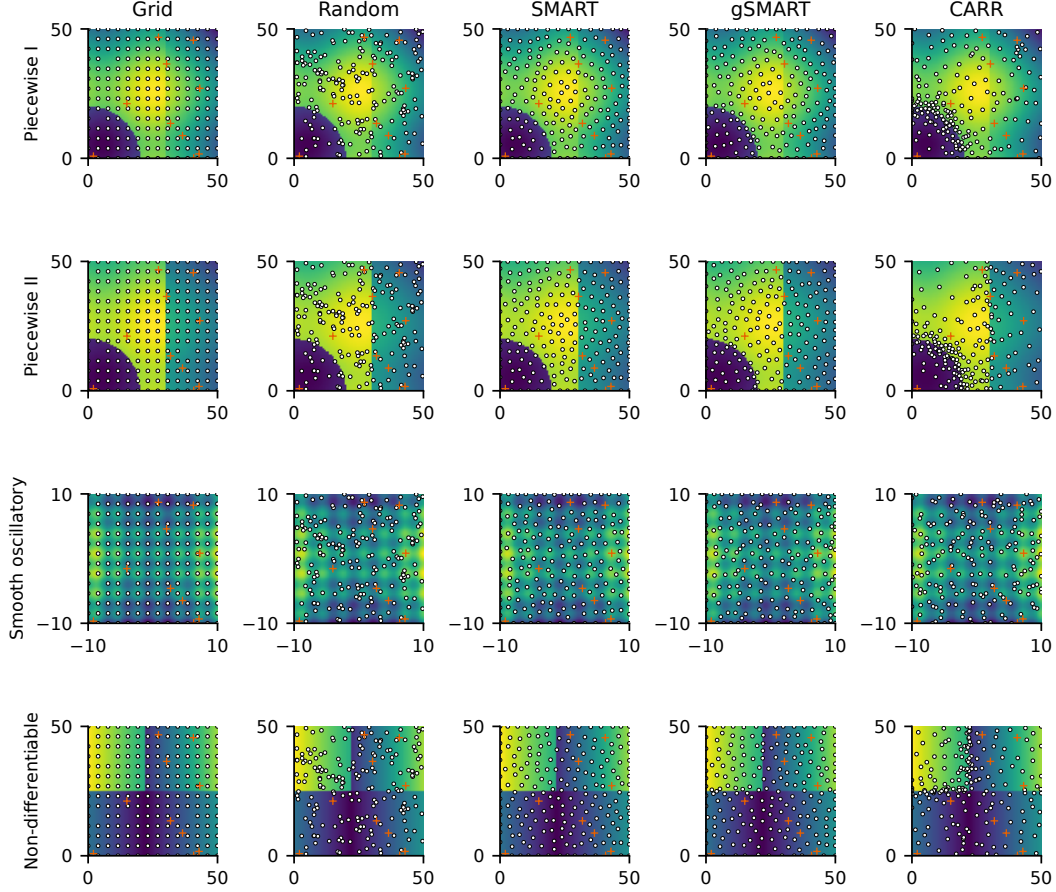


Figure 4: Measurement locations at $n = 200$, seed 0, over the true functions. Rows are the four two-dimensional functions and columns are sampling methods. Orange crosses mark the shared initial design; subsequent measurements are white dots. Colors show the true response within each row. Boundary-focused sampling emerges on the discontinuous functions, while the smooth oscillatory case retains distributed coverage.

If their means are close. Thus the counterfactual mechanism is supported where the error guide is informative; the audit does not validate it uniformly, nor does it test changes caused by refitting the variogram. Figure 9 in the supplement shows the underlying score–gain relation, including negative realized gains.

5.2 Failure analysis: accuracy can inflate a misspecified uncertainty model

The most consequential failure is probabilistic rather than geometric. A stationary exponential model cannot explain arbitrarily close observations with different response levels without increasing its roughness or amplitude. This is visible directly in the fitted global process scale.

Proposition 2 (A sampled jump constrains the fitted scale). *For the ordinary-Kriging estimate in Eq. (2) and any two observations $i \neq j$,*

$$\hat{s}^2 \geq \frac{(y_i - y_j)^2}{2(n-1)\{1 - e^{-\hat{a}\|x_i - x_j\|^2} + \nu\}}. \quad (13)$$

Proof. Set $r = y - \hat{\beta}\mathbf{1}$ and $d = e_i - e_j$. Cauchy–Schwarz in the R inner product gives $(d^T r)^2 \leq (r^T R^{-1} r)(d^T R d)$. Since $d^T r = y_i - y_j$ and $d^T R d = 2(1 - k(x_i, x_j) + \nu)$, division by $n - 1$ yields the bound. The numerical lower floor on $\hat{s}^2$ cannot invalidate it. $\square$

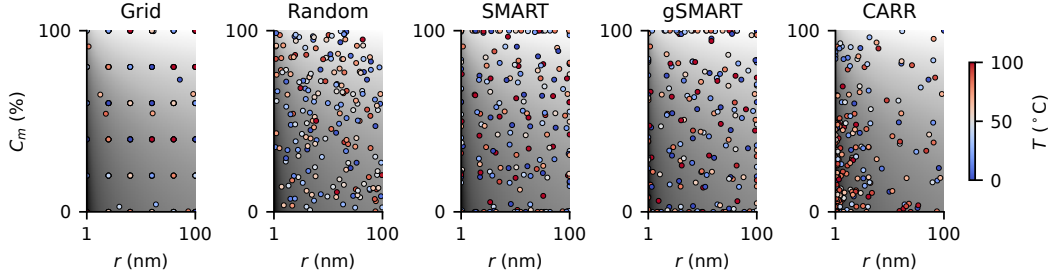


Figure 5: Diffusion measurement locations at $n = 200$, seed 0, projected onto radius and concentration. Point color gives the measured temperature. The grayscale background is the true diffusion function at $T = 50^\circ\text{C}$ on a logarithmic scale (dark indicates higher response); projected points need not lie in this slice. CARR directs more observations toward the small-radius region while preserving interior coverage.

Table 6: Serial decision cost in milliseconds on Piecewise II, averaging decisions at $n = 190, \dots, 199$ within each of ten seeds, then reporting mean $\pm$ seed standard deviation. Guide cost is amortized across its ten-decision refresh cycle; acquisition includes candidates, quadrature, guide prediction, and gpCAM maximization. Function evaluation and diagnostic test-grid prediction are excluded.

Method	GP fit	Guide fit	Acquisition	Total
Grid	2.85 ± 0.17	0.00 ± 0.00	0.16 ± 0.01	3.01 ± 0.17
Random	2.81 ± 0.15	0.00 ± 0.00	0.05 ± 0.00	2.87 ± 0.15
SMART	2.72 ± 0.16	0.00 ± 0.00	3.94 ± 0.73	6.65 ± 0.77
gSMART	2.52 ± 0.20	0.00 ± 0.00	8.36 ± 0.50	10.88 ± 0.68
CARR	2.24 ± 0.09	16.34 ± 3.85	57.23 ± 34.74	75.81 ± 35.77

For a fixed nonzero jump, decreasing separation tightens this lower bound until the nugget dominates. It does not prove that every new observation increases the fitted scale, because n and $\hat{a}$ also change. The bound is checked for every final design in the diagnostic logs. CARR’s mean fitted scale is 3.3 times that of gSMART on Piecewise I, 2.5 times on Piecewise II, and 9.4 times on the piecewise-linear case. The associated CRPS increases and overcoverage show why an acquisition correction alone is insufficient for reliable uncertainty. CARR’s local Q can guide useful sampling while the returned global C remains too diffuse.

Early-budget failures have a separate cause. On all three discontinuous functions the guide can misplace a branch or assign an innovation to the wrong region, and the domain integral spreads that error. The 50-observation regressions are therefore consistent with the same transport mechanism that helps later. On oscillations, a structured additive guide helps but the rough, continuous Kriging predictor still needs broad coverage. On the piecewise-linear case, knowing affine interiors does not remove the need to resolve the split. These properties explain why lower final error does not translate into an observed twofold saving on those four functions.

Diffusion highlights two further limitations. First, the absolute-loss objective underweights relative error at very small responses; the MAPE regression is not repaired by better estimates of signed absolute error. Second, a large efficiency ratio against face-seeking isotropic baselines partly reflects baseline misspecification. Anisotropic kernels and input-dependent observation models are established alternatives in autonomous experimentation [24], but changing them here would confound the acquisition comparison. We have not shown that the same ratio survives a stronger, well-calibrated surrogate.

5.3 Computational cost trade-off

Table 6 gives serial replay timings on Piecewise II for ten decisions per seed at approximately 200 observations. CARR takes 75.8 ms per decision including GP fitting, compared with 6.7 ms for SMART and 10.9 ms for gSMART: factors of 11.4 and 7.0, respectively. Variation includes different guide-prediction workloads across learned weights. These measurements isolate algorithm time from

parallel-worker contention and exclude dense diagnostic prediction. They are not beamline latency measurements.

The dense GP factorization scales cubically with n . After fitting, acquisition covariance work is dominated by matrix products linking n observations, M integration points, and the candidate pool; the M -by-candidate loss matrix adds linear work in their product. The reference experts add periodic training and prediction cost. At the measured problem size, the absolute overhead is a fraction of a second, potentially acceptable when physical measurements take much longer. It is wasteful when the oracle is as cheap as these analytic functions. More importantly, where CARR's gain over a simple disagreement rule is absent, its extra computational complexity lacks empirical justification. The decision-time evidence supports feasibility for expensive measurements, not universal cost effectiveness.

5.4 Limits of the evidence

The five analytic functions cover several useful structures but are too few to establish broad superiority in adaptive design. Development used this same benchmark, and the guide library contains additive and partition-linear models that fit two functions particularly well. Seeds share initial designs across methods, but independent seeds do not remove this family-selection bias. The residual scale fit reuses stacked residuals and conditionally fixed hyperparameters; it is neither a fully Bayesian error model nor a calibrated posterior. Candidate-pool maximization approximates a continuous design problem, and the finite resolution of the test quadrature can affect estimates near sharp transitions. No noisy observations, instrument constraints, travel costs, or high-dimensional problems were tested. Most directly, the current study neither establishes twofold measurement savings on four functions nor fully resolves the original baseline ranking. Those limits constrain the acquisition claim even though every reported final sweep is complete.

6 Conclusion

Signed error transport offers a direct way to ask whether a measurement will correct the reconstruction that is actually returned. Its usefulness depends on two learnable quantities: the direction of the surrogate's error and the spatial distribution of residual uncertainty. With a common Kriging predictor, CARR improves final absolute error across the five studied functions, but substantial measurement savings are established only for diffusion under the implemented baseline configuration. Boundary recovery is improved primarily by spatial error scaling; retaining the sign is not uniformly beneficial. The same sampling decisions can worsen probabilistic prediction by forcing a stationary model to explain short-distance jumps with a global variance. Acquisition design should therefore be assessed against both its reconstruction objective and the calibration of the model it leaves behind. Extending the error law to a nonstationary predictive model is a promising next step, but the present evidence does not establish that extension or broad superiority over modern adaptive-design methods.

Reproducibility and scope. The accompanying artifact contains the source, complete trajectories, per-refresh fits, dense-grid predictions, metric aggregates, and scripts producing every table and figure. The execution and unsuccessful alternatives are documented in `codebase_notes.md` and `attempts_log.md`. All results concern noiseless analytic functions; no instrument data or beamline experiment is represented as having been collected.

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Supplementary numerical specification and results

This supplement gives exact benchmark definitions and complete predictive-quality results supporting the main text. The full per-seed values, including coverage at all nominal levels for every control and budget, are in `logs/all_metrics.csv`. Aggregates, including sample standard deviations, are in `logs/aggregates.json`. These are generated from completed trajectories rather than transcribed from figures.

Benchmark equations

For $(x, y) \in [0, 50]^2$, let $B(x, y) = 8 - 2x \sin \sqrt{|x|} - 2y \sin \sqrt{|y|}$. The two piecewise-sinusoidal functions are

$$f_j(x, y) = B(x, y) + \begin{cases} a_j, & x \geq 30, \\ b_j, & x < 30 \text{ and } \sqrt{x^2 + y^2} \geq 20, \\ 0, & \text{otherwise,} \end{cases} \quad (14)$$

Here $(a_1, b_1) = (140, 160)$ and $(a_2, b_2) = (200, 360)$. The order of conditions matters: the $x \geq 30$ branch overrides the radial condition. The piecewise-linear function on the same square is

$$f(x, y) = \begin{cases} \frac{25}{12}(x - 5) + 14, & x > 21.961143, y > 25, \\ \frac{25}{17}(x - 5) + 14, & x > 21.961143, y \leq 25, \\ -\frac{25}{18}(x - 50) + 50, & x \leq 21.961143, y > 25, \\ -\frac{25}{18}(x - 50), & x \leq 21.961143, y \leq 25. \end{cases} \quad (15)$$

These implement Eqs. 12, 13, and 17 of Noack et al. [22]. The smooth function is given explicitly in the main text. Inputs are transformed affinely into the unit cube for every fit and acquisition.

For diffusion, let $c = C_m/100$ and use Celsius temperature T in the viscosity expressions of Cheng [5]:

$$\mu_w = 1.790 \exp\left(\frac{(-1230 - T)T}{36100 + 360T}\right), \quad \mu_g = 12100 \exp\left(\frac{(-1233 + T)T}{9900 + 70T}\right), \quad (16)$$

$$a_v = 0.705 - 0.0017T, \quad b_v = (4.9 + 0.036T)a_v^{2.5}, \quad (17)$$

$$\alpha_v = 1 - c + \frac{a_v b_v c(1 - c)}{a_v c + b_v(1 - c)}, \quad \mu = 10^{-3} \mu_w^{\alpha_v} \mu_g^{1 - \alpha_v} \text{ Pa s.} \quad (18)$$

The viscosities μ_w, μ_g are in mPa s before conversion. The reported coefficient is

$$D(r, T, C_m) = 10^{12} \frac{1.380649 \times 10^{-23}(T + 273.15)}{6\pi\mu(r \times 10^{-9})} \mu\text{m}^2/\text{s}, \quad (19)$$

with r expressed in nm. No simplified exponential mixture approximation is used.

Predictive scoring and fitted guide behavior

For Gaussian mean m , standard deviation s , response y , and $t = (y - m)/s$, the scored distribution has

$$\text{CRPS}(m, s; y) = s \left[t(2\Phi(t) - 1) + 2\phi(t) - \frac{1}{\sqrt{\pi}} \right]. \quad (20)$$

We average this expression over the independent evaluation grid. Central coverage at level p is the fraction satisfying $|y - m| \leq \Phi^{-1}((1 + p)/2)s$. These scores use the unchanged Kriging variance $v = s^2$, not the acquisition error covariance Q . The latter is only a decision aid and has no probabilistic-forecast performance claim.

Table 7: Fitted guide behavior at the last refresh ($n = 490$ for 2D, 990 for diffusion): mean simplex weights across ten seeds, median global scale $\bar{\lambda}$, and mean spatial-shrinkage weight η . GP, TPS, ET, AS, PL, and AT denote Kriging, thin-plate spline, extra trees, additive spline, partition-linear, and asinh thin-plate experts. Rows use the identical fitting procedure.

Function	GP	TPS	ET	AS	PL	AT	$\bar{\lambda}$	η
Piecewise I	0.579	0.152	0.212	0.004	0.030	0.023	0.9047	1.000
Piecewise II	0.158	0.102	0.642	0.000	0.065	0.032	0.9842	1.000
Smooth osc.	0.000	0.000	0.000	0.999	0.000	0.001	0.0004	0.175
Non-diff.	0.000	0.000	0.107	0.003	0.890	0.000	0.0894	0.275
3D diffusion	0.000	0.000	0.697	0.000	0.288	0.015	0.0425	1.000

Table 8: Predictive quality on every headline configuration. Entries are mean $\pm$ standard deviation over ten seeds. Coverage is in percent. Units match Table 1.

Function	Method	n	RMSE	CRPS	Coverage 90%	Coverage 95%
Piecewise I	Grid	50	19.555 $\pm$ 1.309	9.941 $\pm$ 0.349	94.8 $\pm$ 0.7	96.0 $\pm$ 0.6
	Random	50	24.502 $\pm$ 3.367	11.881 $\pm$ 1.077	94.0 $\pm$ 1.7	95.2 $\pm$ 1.6
	SMART	50	20.024 $\pm$ 1.962	9.829 $\pm$ 0.523	94.6 $\pm$ 0.5	95.7 $\pm$ 0.5
	gSMART	50	17.640 $\pm$ 1.847	10.194 $\pm$ 0.340	96.2 $\pm$ 0.5	97.1 $\pm$ 0.5
	CARR	50	18.823 $\pm$ 1.952	10.820 $\pm$ 0.858	96.2 $\pm$ 0.9	96.9 $\pm$ 0.8
Piecewise I	Grid	100	16.454 $\pm$ 1.104	7.253 $\pm$ 0.251	95.3 $\pm$ 0.4	96.0 $\pm$ 0.4
	Random	100	18.446 $\pm$ 2.209	8.592 $\pm$ 0.758	94.7 $\pm$ 1.6	95.5 $\pm$ 1.5
	SMART	100	15.934 $\pm$ 1.034	7.184 $\pm$ 0.223	95.2 $\pm$ 0.4	95.9 $\pm$ 0.4
	gSMART	100	14.887 $\pm$ 0.693	8.001 $\pm$ 0.160	96.8 $\pm$ 0.2	97.3 $\pm$ 0.1
	CARR	100	13.701 $\pm$ 1.487	9.355 $\pm$ 0.966	97.5 $\pm$ 0.8	97.9 $\pm$ 0.7
Piecewise I	Grid	200	13.629 $\pm$ 0.508	5.644 $\pm$ 0.099	96.1 $\pm$ 0.2	96.5 $\pm$ 0.2
	Random	200	14.670 $\pm$ 1.006	6.389 $\pm$ 0.188	95.8 $\pm$ 0.6	96.4 $\pm$ 0.5
	SMART	200	13.000 $\pm$ 0.784	5.458 $\pm$ 0.178	96.1 $\pm$ 0.2	96.5 $\pm$ 0.2
	gSMART	200	11.714 $\pm$ 0.664	6.299 $\pm$ 0.215	97.5 $\pm$ 0.1	97.8 $\pm$ 0.1
	CARR	200	9.802 $\pm$ 0.665	8.857 $\pm$ 0.951	98.5 $\pm$ 0.3	98.8 $\pm$ 0.3
Piecewise I	Grid	500	10.119 $\pm$ 0.253	3.894 $\pm$ 0.031	97.2 $\pm$ 0.1	97.4 $\pm$ 0.1
	Random	500	11.624 $\pm$ 0.857	4.488 $\pm$ 0.164	96.7 $\pm$ 0.5	97.0 $\pm$ 0.4
	SMART	500	10.250 $\pm$ 0.565	3.963 $\pm$ 0.120	97.2 $\pm$ 0.1	97.4 $\pm$ 0.1
	gSMART	500	8.920 $\pm$ 0.487	4.860 $\pm$ 0.115	98.4 $\pm$ 0.1	98.5 $\pm$ 0.1
	CARR	500	6.661 $\pm$ 0.257	9.593 $\pm$ 0.673	99.3 $\pm$ 0.1	99.4 $\pm$ 0.1
Piecewise II	Grid	50	48.356 $\pm$ 2.769	24.197 $\pm$ 0.805	93.7 $\pm$ 1.1	95.3 $\pm$ 0.8
	Random	50	57.471 $\pm$ 6.793	28.393 $\pm$ 2.515	92.3 $\pm$ 2.6	94.2 $\pm$ 2.2
	SMART	50	50.745 $\pm$ 4.269	24.879 $\pm$ 1.494	92.5 $\pm$ 1.3	94.5 $\pm$ 0.9
	gSMART	50	46.853 $\pm$ 2.880	25.046 $\pm$ 1.119	94.8 $\pm$ 0.4	96.2 $\pm$ 0.4
	CARR	50	47.934 $\pm$ 4.636	26.597 $\pm$ 1.133	95.1 $\pm$ 1.9	96.4 $\pm$ 1.5
Piecewise II	Grid	100	40.396 $\pm$ 2.258	18.322 $\pm$ 0.543	93.8 $\pm$ 0.7	95.4 $\pm$ 0.6
	Random	100	45.239 $\pm$ 4.268	21.688 $\pm$ 1.330	93.0 $\pm$ 2.6	94.6 $\pm$ 2.2
	SMART	100	40.300 $\pm$ 2.287	18.743 $\pm$ 0.532	93.0 $\pm$ 0.8	94.6 $\pm$ 0.8
	gSMART	100	37.617 $\pm$ 1.750	19.749 $\pm$ 0.501	95.6 $\pm$ 0.4	96.7 $\pm$ 0.3
	CARR	100	35.988 $\pm$ 2.460	22.387 $\pm$ 1.783	96.8 $\pm$ 0.6	97.6 $\pm$ 0.5
Piecewise II	Grid	200	35.215 $\pm$ 0.941	15.166 $\pm$ 0.205	93.2 $\pm$ 0.2	94.2 $\pm$ 0.4
	Random	200	37.476 $\pm$ 2.324	16.936 $\pm$ 0.337	93.6 $\pm$ 1.1	94.9 $\pm$ 1.0
	SMART	200	33.769 $\pm$ 1.310	14.774 $\pm$ 0.254	93.6 $\pm$ 0.5	94.8 $\pm$ 0.5
	gSMART	200	30.172 $\pm$ 1.045	16.009 $\pm$ 0.394	96.4 $\pm$ 0.4	97.2 $\pm$ 0.3
	CARR	200	26.170 $\pm$ 1.410	21.918 $\pm$ 2.705	98.3 $\pm$ 0.5	98.7 $\pm$ 0.4
Piecewise II	Grid	500	26.865 $\pm$ 0.499	10.806 $\pm$ 0.083	94.6 $\pm$ 0.3	95.5 $\pm$ 0.2
	Random	500	29.579 $\pm$ 1.519	12.314 $\pm$ 0.319	94.7 $\pm$ 0.6	95.6 $\pm$ 0.5
	SMART	500	26.038 $\pm$ 1.533	10.666 $\pm$ 0.303	95.1 $\pm$ 0.2	95.8 $\pm$ 0.2
	gSMART	500	23.079 $\pm$ 1.020	12.550 $\pm$ 0.215	97.3 $\pm$ 0.2	97.9 $\pm$ 0.2
	CARR	500	17.431 $\pm$ 0.664	21.696 $\pm$ 1.618	99.2 $\pm$ 0.1	99.3 $\pm$ 0.1
Smooth osc.	Grid	50	1.257 $\pm$ 0.032	0.736 $\pm$ 0.023	70.3 $\pm$ 4.8	79.6 $\pm$ 4.6
	Random	50	1.191 $\pm$ 0.023	0.671 $\pm$ 0.013	85.0 $\pm$ 3.3	92.4 $\pm$ 2.5
	SMART	50	1.277 $\pm$ 0.067	0.745 $\pm$ 0.045	73.5 $\pm$ 4.1	82.6 $\pm$ 3.7
	gSMART	50	1.240 $\pm$ 0.049	0.716 $\pm$ 0.032	77.5 $\pm$ 3.7	86.1 $\pm$ 3.5
	CARR	50	1.172 $\pm$ 0.047	0.663 $\pm$ 0.031	84.4 $\pm$ 4.2	91.9 $\pm$ 3.4

Function	Method	n	RMSE	CRPS	Coverage 90%	Coverage 95%
Smooth osc.	Grid	100	1.199 ± 0.015	0.711 ± 0.016	66.4 ± 3.2	76.2 ± 3.2
	Random	100	1.074 ± 0.025	0.595 ± 0.015	86.4 ± 2.5	93.4 ± 1.8
	SMART	100	1.089 ± 0.035	0.615 ± 0.025	82.6 ± 4.5	90.8 ± 3.5
	gSMART	100	1.093 ± 0.028	0.619 ± 0.017	80.4 ± 2.6	89.1 ± 2.3
	CARR	100	0.927 ± 0.026	0.517 ± 0.012	96.3 ± 3.4	99.2 ± 1.5
Smooth osc.	Grid	200	0.437 ± 0.024	0.319 ± 0.004	99.9 ± 0.1	100.0 ± 0.0
	Random	200	0.889 ± 0.020	0.476 ± 0.013	89.2 ± 2.2	94.9 ± 1.4
	SMART	200	0.687 ± 0.015	0.386 ± 0.006	98.4 ± 0.6	99.8 ± 0.2
	gSMART	200	0.753 ± 0.028	0.412 ± 0.012	95.6 ± 1.3	99.0 ± 0.4
	CARR	200	0.445 ± 0.021	0.326 ± 0.003	100.0 ± 0.0	100.0 ± 0.0
Smooth osc.	Grid	500	0.262 ± 0.008	0.178 ± 0.002	99.9 ± 0.1	100.0 ± 0.0
	Random	500	0.500 ± 0.034	0.259 ± 0.013	96.1 ± 1.3	98.5 ± 0.8
	SMART	500	0.247 ± 0.005	0.176 ± 0.001	100.0 ± 0.0	100.0 ± 0.0
	gSMART	500	0.319 ± 0.010	0.195 ± 0.003	100.0 ± 0.0	100.0 ± 0.0
	CARR	500	0.129 ± 0.003	0.169 ± 0.000	100.0 ± 0.0	100.0 ± 0.0
Non-diff.	Grid	50	9.286 ± 0.784	4.171 ± 0.222	89.2 ± 2.4	91.0 ± 2.3
	Random	50	8.310 ± 0.959	4.011 ± 0.293	90.5 ± 3.8	92.3 ± 3.4
	SMART	50	7.271 ± 0.812	3.486 ± 0.263	91.9 ± 1.4	93.7 ± 1.4
	gSMART	50	6.621 ± 0.494	3.472 ± 0.175	94.1 ± 0.7	95.5 ± 0.7
	CARR	50	6.682 ± 0.563	3.785 ± 0.353	95.1 ± 2.0	96.6 ± 1.6
Non-diff.	Grid	100	6.101 ± 0.358	2.780 ± 0.121	93.2 ± 1.0	94.4 ± 1.0
	Random	100	6.272 ± 0.399	2.999 ± 0.210	92.9 ± 1.8	94.4 ± 1.6
	SMART	100	5.986 ± 0.479	2.672 ± 0.162	92.8 ± 0.3	94.0 ± 0.3
	gSMART	100	5.287 ± 0.250	2.725 ± 0.065	95.0 ± 0.5	96.1 ± 0.5
	CARR	100	4.290 ± 0.383	3.445 ± 0.575	97.9 ± 1.1	98.5 ± 0.9
Non-diff.	Grid	200	4.012 ± 0.044	1.828 ± 0.022	94.3 ± 0.3	94.9 ± 0.3
	Random	200	5.182 ± 0.378	2.339 ± 0.113	93.7 ± 1.2	94.8 ± 1.1
	SMART	200	4.906 ± 0.251	2.086 ± 0.072	93.9 ± 0.3	94.9 ± 0.3
	gSMART	200	4.450 ± 0.182	2.240 ± 0.042	95.9 ± 0.4	96.7 ± 0.3
	CARR	200	3.050 ± 0.177	4.073 ± 0.594	99.3 ± 0.3	99.5 ± 0.2
Non-diff.	Grid	500	4.800 ± 0.084	1.637 ± 0.016	94.8 ± 0.4	95.2 ± 0.2
	Random	500	4.200 ± 0.238	1.700 ± 0.077	94.7 ± 0.6	95.5 ± 0.6
	SMART	500	3.685 ± 0.133	1.501 ± 0.030	95.4 ± 0.2	96.0 ± 0.1
	gSMART	500	3.258 ± 0.142	1.728 ± 0.063	97.2 ± 0.2	97.7 ± 0.2
	CARR	500	1.903 ± 0.082	4.524 ± 0.435	99.8 ± 0.1	99.9 ± 0.0
3D diffusion	Grid	50	65.999 ± 9.062	30.958 ± 2.178	94.3 ± 1.7	95.4 ± 1.5
	Random	50	17.458 ± 1.510	4.819 ± 1.320	93.9 ± 2.3	94.9 ± 2.0
	SMART	50	50.636 ± 9.690	26.819 ± 2.358	96.9 ± 1.4	97.5 ± 1.2
	gSMART	50	33.810 ± 6.138	29.467 ± 1.436	99.0 ± 0.5	99.3 ± 0.4
	CARR	50	15.914 ± 4.584	18.159 ± 9.003	99.2 ± 0.6	99.4 ± 0.5
3D diffusion	Grid	100	49.873 ± 3.468	21.392 ± 1.987	93.5 ± 1.6	94.5 ± 1.4
	Random	100	15.845 ± 1.826	4.380 ± 1.187	94.9 ± 2.1	95.7 ± 1.9
	SMART	100	39.130 ± 6.467	19.698 ± 0.884	97.1 ± 0.8	97.7 ± 0.7
	gSMART	100	26.660 ± 4.562	21.959 ± 0.872	98.9 ± 0.4	99.3 ± 0.4
	CARR	100	10.800 ± 1.794	18.776 ± 5.753	99.7 ± 0.3	99.8 ± 0.2
3D diffusion	Grid	200	43.495 ± 2.105	17.219 ± 0.820	94.6 ± 0.5	95.3 ± 0.4
	Random	200	14.266 ± 1.616	4.061 ± 0.915	95.8 ± 1.5	96.5 ± 1.3
	SMART	200	31.617 ± 3.928	14.966 ± 0.695	96.9 ± 0.4	97.5 ± 0.4
	gSMART	200	24.635 ± 2.509	18.201 ± 1.026	98.5 ± 0.3	98.9 ± 0.2
	CARR	200	7.692 ± 0.842	21.884 ± 4.206	99.9 ± 0.1	99.9 ± 0.1
3D diffusion	Grid	500	33.595 ± 0.382	12.099 ± 0.055	95.2 ± 0.2	95.8 ± 0.2
	Random	500	11.988 ± 1.651	3.416 ± 0.839	96.7 ± 1.3	97.2 ± 1.1
	SMART	500	25.804 ± 2.359	10.985 ± 0.302	96.8 ± 0.1	97.3 ± 0.1
	gSMART	500	18.929 ± 1.024	13.558 ± 0.566	98.4 ± 0.3	98.8 ± 0.2
	CARR	500	6.071 ± 0.396	23.059 ± 0.696	99.9 ± 0.0	100.0 ± 0.0
3D diffusion	Grid	1000	27.039 ± 0.241	9.186 ± 0.026	95.6 ± 0.1	96.2 ± 0.1
	Random	1000	9.920 ± 0.937	2.979 ± 0.412	97.4 ± 0.7	97.8 ± 0.6
	SMART	1000	21.206 ± 2.029	8.693 ± 0.183	97.0 ± 0.3	97.4 ± 0.2
	gSMART	1000	16.562 ± 0.703	11.147 ± 0.216	98.1 ± 0.2	98.6 ± 0.2
	CARR	1000	5.171 ± 0.211	20.897 ± 0.774	99.9 ± 0.0	99.9 ± 0.0

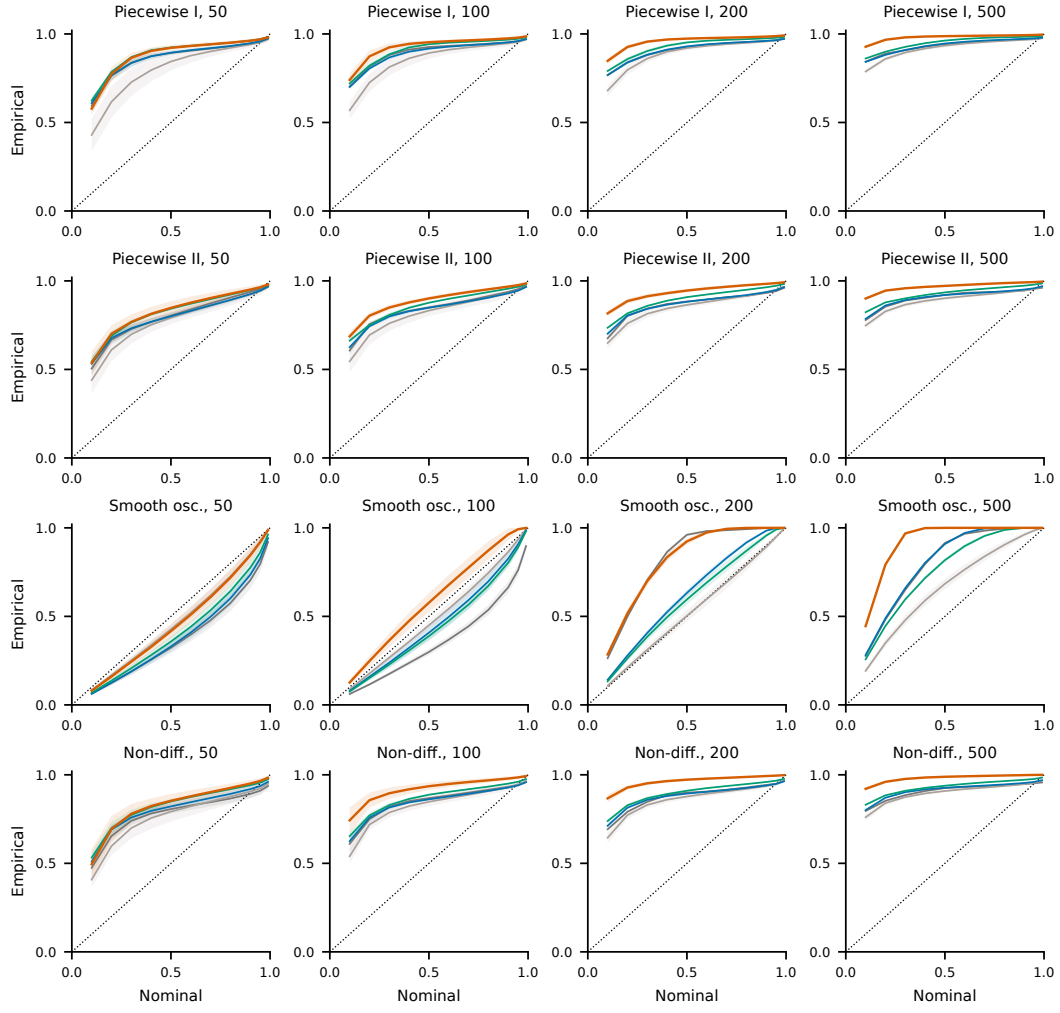


Figure 6: Complete two-dimensional probability-coverage curves at all budgets, with ten-seed means and one-standard-deviation bands. Columns increase in budget. Colors match Figure 2: gray grid, taupe random, blue SMART, green gSMART, orange CARR.

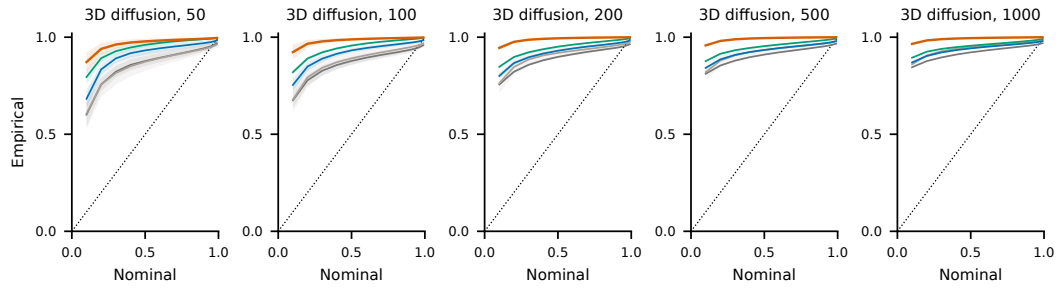


Figure 7: Complete diffusion coverage curves, at budgets 50, 100, 200, 500, and 1000 from left to right. Colors and bands follow Figure 6.

Table 9: Predictive metrics for acquisition controls at each function's maximum budget; ten seeds. Coverage is percent.

Function	Control	RMSE	CRPS	Coverage 90%	Coverage 95%
Piecewise I	Zero bias	6.332 ± 0.275	9.961 ± 0.717	99.3 ± 0.1	99.5 ± 0.1
	Unsigned bias	6.469 ± 0.379	9.537 ± 0.941	99.3 ± 0.1	99.4 ± 0.1
	Global scale	9.820 ± 0.853	4.548 ± 0.401	97.8 ± 0.4	98.0 ± 0.4
	Disagreement	6.104 ± 0.376	13.102 ± 1.619	99.6 ± 0.1	99.7 ± 0.1
	ISR	10.572 ± 0.761	4.026 ± 0.164	97.1 ± 0.2	97.4 ± 0.2
	IVR	10.588 ± 0.474	3.988 ± 0.132	97.1 ± 0.2	97.4 ± 0.2
	EIGF	6.996 ± 0.300	18.505 ± 0.878	99.8 ± 0.0	99.9 ± 0.0
Piecewise II	Zero bias	18.397 ± 0.661	22.374 ± 1.594	99.0 ± 0.1	99.3 ± 0.1
	Unsigned bias	17.504 ± 0.540	21.585 ± 1.463	99.1 ± 0.1	99.3 ± 0.1
	Global scale	23.896 ± 1.892	12.444 ± 0.935	97.1 ± 0.8	97.6 ± 0.7
	Disagreement	15.558 ± 0.631	35.916 ± 3.341	99.6 ± 0.1	99.7 ± 0.1
	ISR	26.679 ± 1.518	10.891 ± 0.296	95.2 ± 0.3	95.9 ± 0.3
	IVR	27.246 ± 1.124	10.914 ± 0.265	95.0 ± 0.3	95.8 ± 0.4
	EIGF	20.326 ± 0.647	43.136 ± 2.310	99.9 ± 0.0	99.9 ± 0.0
Smooth osc.	Zero bias	0.300 ± 0.015	0.189 ± 0.004	100.0 ± 0.1	100.0 ± 0.0
	Unsigned bias	0.126 ± 0.002	0.166 ± 0.000	100.0 ± 0.0	100.0 ± 0.0
	Global scale	0.127 ± 0.002	0.168 ± 0.000	100.0 ± 0.0	100.0 ± 0.0
	Disagreement	0.116 ± 0.002	0.165 ± 0.000	100.0 ± 0.0	100.0 ± 0.0
	ISR	0.293 ± 0.005	0.187 ± 0.001	100.0 ± 0.0	100.0 ± 0.0
	IVR	0.279 ± 0.008	0.183 ± 0.002	100.0 ± 0.0	100.0 ± 0.0
	EIGF	0.557 ± 0.025	0.279 ± 0.009	93.6 ± 0.9	96.9 ± 0.6
Non-diff.	Zero bias	3.289 ± 0.429	2.042 ± 0.373	97.2 ± 1.0	97.7 ± 1.0
	Unsigned bias	1.844 ± 0.098	5.513 ± 0.694	99.9 ± 0.1	100.0 ± 0.0
	Global scale	2.013 ± 0.136	4.267 ± 0.565	99.8 ± 0.1	99.9 ± 0.1
	Disagreement	1.859 ± 0.030	14.853 ± 0.536	100.0 ± 0.0	100.0 ± 0.0
	ISR	3.826 ± 0.247	1.566 ± 0.038	95.4 ± 0.5	96.0 ± 0.5
	IVR	3.895 ± 0.110	1.559 ± 0.041	95.3 ± 0.2	95.9 ± 0.2
	EIGF	2.648 ± 0.061	5.609 ± 0.222	99.9 ± 0.0	100.0 ± 0.0
3D diffusion	Zero bias	8.714 ± 0.583	14.565 ± 0.643	99.5 ± 0.1	99.6 ± 0.0
	Unsigned bias	5.925 ± 0.133	22.946 ± 0.572	99.9 ± 0.0	99.9 ± 0.0
	Global scale	5.923 ± 0.360	15.925 ± 0.645	99.9 ± 0.0	99.9 ± 0.0
	Disagreement	6.076 ± 0.076	41.567 ± 1.214	100.0 ± 0.0	100.0 ± 0.0
	ISR	13.901 ± 2.558	6.493 ± 0.540	98.1 ± 0.3	98.4 ± 0.3
	IVR	14.295 ± 1.487	6.632 ± 0.308	98.0 ± 0.5	98.3 ± 0.4
	EIGF	8.972 ± 0.213	30.998 ± 0.387	100.0 ± 0.0	100.0 ± 0.0

Table 10: Diffusion MAPE for every acquisition control at 1000 observations (percent, ten-seed mean $\pm$ standard deviation).

Acquisition	MAPE (%)
CARR	796.7 ± 150.5
Zero bias	2563.7 ± 1049.7
Unsigned bias	1181.8 ± 306.0
Global scale	538.9 ± 146.2
Disagreement	4431.5 ± 1343.7
ISR	229.3 ± 41.5
IVR	211.2 ± 60.1
EIGF	446.3 ± 44.4

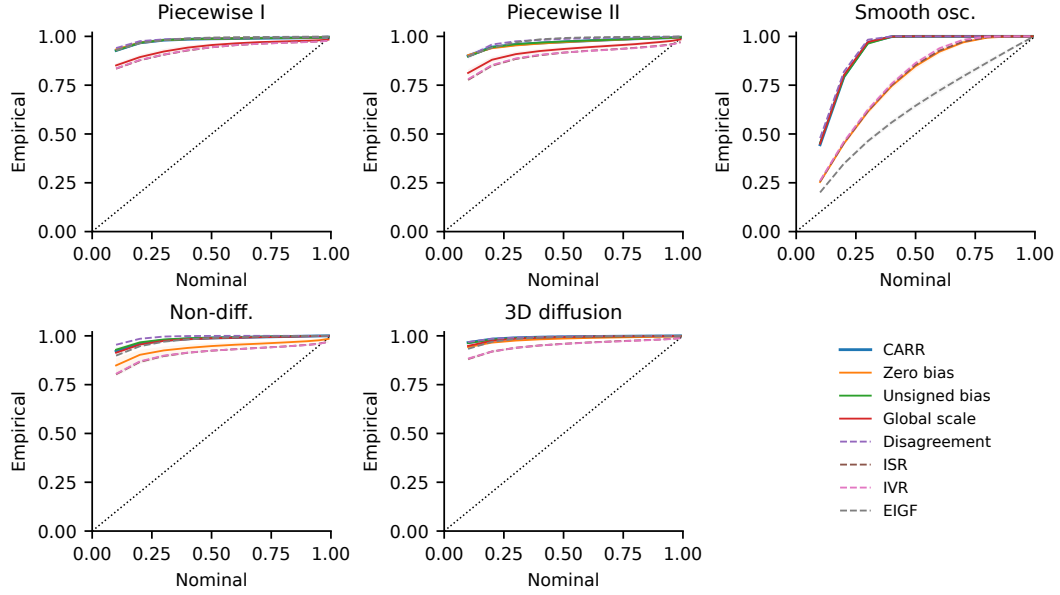


Figure 8: Probability-coverage curves for all acquisition controls at their reported maximum budgets, with ten-seed means and one-standard-deviation bands. The simple and integrated-risk controls share the same predictive distribution formula; sampling changes its fitted parameters and coverage.

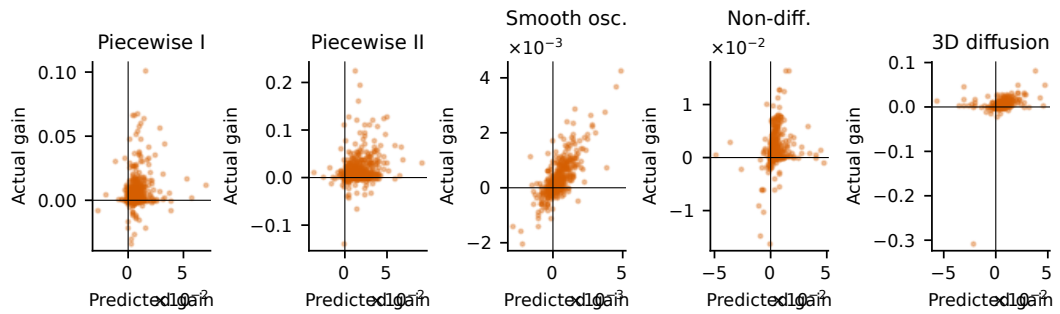


Figure 9: CARR's predicted absolute-error reduction against the realized fixed-kernel reduction at 64 independent audit candidates for each of ten histories per function ($n = 200$). Points from the ten seeds are pooled only for visualization; Table 5 averages within-seed rank correlations. Both the realized gain and the model-based score can be negative. Extra oracle calls for this audit never affect sequential trajectories.

A Unresolved aspects of baseline reproduction

The original SMART study [22] motivates variance and gradient-weighted acquisition, but not every reported qualitative ranking is recovered under the common-predictor protocol used here. The main baseline sweep is complete: SMART, normalized gSMART, random, and grid each have ten seeds at every requested function and budget. Mean error decreases with budget for every baseline curve. However, SMART and gSMART do not consistently dominate full-domain grids in two dimensions and are substantially worse than random sampling for diffusion. We retain these results rather than substituting values from the original publication.

Several distinct configuration checks were completed before fixing the headline comparison. First, the exponential kernel and unknown-mean correction were verified algebraically against ordinary Kriging; the default gpCAM kernel and observation noise were not used. Initial finite-difference gradients were replaced with analytic derivatives because clustered points amplified cancellation. Second, an exploratory fixed standardized-gradient coefficient $c = 0.1$ produced severe point concentration near jumps. The candidate-normalized coefficient in Eq. (12) reduced this pathology without function-specific tuning. Third, replacing the squared-range normalizer in Eq. (1) by twice the sample variance was tested on all functions. It changed the fitted scale and some rankings but did not recover the missing qualitative behavior across the benchmark. Fourth, gpCAM’s continuous differential-evolution optimizer was run for SMART and gSMART using population multiplier 10, maximum 20 iterations, and tolerance 10^{-6} . Those runs reduced some candidate-discretization concerns but likewise did not establish universal dominance over grid and random.

These development checks use two seeds and are diagnostics, not substitutes for the ten-seed headline sweep. Their complete trajectories and metrics are stored under `pilot`, `pilot_bounded`, `pilot_sill`, and `optimizer_check`; the exact configurations are in each suite manifest and the outcomes are summarized in `development_summary.json`. The original paper also used interpolation and scan conventions that are not identical to holding ordinary Kriging fixed for every method and using a full-domain lattice at each budget. These differences are plausible contributors, not a demonstrated causal explanation of the residual discrepancy.

Consequently, the headline comparison establishes behavior of the explicitly stated implementations, not a fully validated replication of the original relative ranking. In particular, the diffusion efficiency ratio should be treated cautiously because its reference baselines are weak compared with random sampling. Further work with an independently validated original implementation and stronger surrogate alternatives is needed to remove this limitation. The artifact’s attempts log records the completed checks and the remaining unmet performance target; neither is presented as a successful reproduction.